

Rowing Data Standard for Garmin FIT Format

Version: 1.1

Date: May 14, 2026

Application ID: 74c89dce-0f16-5d5a-aeb1-6c9f3e6d8f3a (UUID v5 from DNS:rowingdata)

1. Introduction

1.1 Purpose

This document defines a standard for encoding rowing-specific data within the Garmin FIT (Flexible and Interoperable Data Transfer) format. The standard enables interoperability between rowing devices, software platforms, and data analysis tools through consistent use of FIT developer fields.

1.2 Scope

This standard specifies:

- Developer field definitions for rowing-specific metrics
- Recording strategies for different device types
- Requirements for data consumers
- In-stroke curve data encoding
- Native FIT field usage for rowing

1.3 Target Audience

- Rowing device manufacturers (ergometers, smart oarlocks, GPS devices)
- Software developers (training platforms, analysis tools)
- Data platform operators

1.4 Application ID

All developer fields defined in this standard **MUST** use the **rowing-data application ID** to prevent namespace collisions.

Format: 16-byte UUID array as required by FIT SDK `DeveloperDataIdMessage`

UUID: 74c89dce-0f16-5d5a-aeb1-6c9f3e6d8f3a

This UUID v5 is deterministically generated from DNS namespace with name “rowingdata”, ensuring consistency across implementa-

tions. The FIT SDK requires the `application_id` field to be a 16-byte array representation of this UUID.

1.5 Conformance Language

The key words “MUST”, “MUST NOT”, “REQUIRED”, “SHALL”, “SHALL NOT”, “SHOULD”, “SHOULD NOT”, “RECOMMENDED”, “MAY”, and “OPTIONAL” in this document are to be interpreted as described in RFC 2119.

2. Recording Strategies

2.1 Overview

Different device types generate FIT Record messages at different frequencies. Consumers **MUST** support both approaches:

2.2 Stroke-Boundary Recording

Definition: One Record message per completed stroke cycle.

Characteristics: - Each Record represents a complete stroke - Stroke-specific metrics (drive time, force, angles) describe that specific stroke - GPS position **MAY** be interpolated or repeated across strokes - **REQUIRED** when in-stroke curve data is present

Typical devices: Indoor ergometers, instrumented oarlocks, on-water measurement systems with stroke detection, rowing-specific data loggers

2.3 GPS-Update Recording

Definition: Record messages generated at GPS position updates.

Characteristics: - Records generated when GPS data is available (typically ~1 Hz, but irregular) - Stroke metrics **MAY** be averaged, interpolated, or represent the most recent stroke - GPS position is precisely measured at record time - **CANNOT** include in-stroke curve data (curves require stroke boundaries) - `total_cycles` field **MAY** repeat across multiple records (same stroke) or skip values (multiple strokes between GPS updates)

Typical devices: GPS-enabled sports watches, multi-sport fitness devices, smartphone applications with GPS tracking

2.4 RecordingStrategy Metadata Field

To indicate the recording approach, producers MAY include the **RecordingStrategy** developer field (ID 10, UINT8) on the **Session message**.

Value	Name	Description
0	Unknown	Recording strategy unspecified (consumers MUST handle both approaches)
1	StrokeBoundary	One Record per stroke cycle
2	GPSUpdate	Records at GPS position updates

Message Type: Session (one value per file)

Base Type: UINT8

Scale: 1

Units: (none)

When omitted or zero, consumers MUST NOT assume any particular strategy.

3. Consumer Requirements

3.1 Mandatory Requirements

Consumers MUST:

1. **Support both recording strategies** without requiring configuration
2. **Not assume 1:1 correspondence** between Record messages and strokes
3. **Not interpolate stroke-specific developer fields** between records (DriveLength, StrokeDriveTime, Catch, Finish, oarlock angles, etc.) - these describe discrete stroke events
4. **Detect stroke occurrences** by monitoring changes in the native `total_cycles` field:
 - When `total_cycles` changes between consecutive records, at least one stroke occurred
 - If change is >1 , multiple strokes occurred but per-stroke data for intermediate strokes is unavailable
5. **Calculate stroke rate** from the native cadence field (strokes/min), not from record message frequency
6. **Handle missing developer fields gracefully** (all developer fields are optional)

3.2 Recommended Requirements

Consumers SHOULD:

1. Read RecordingStrategy from Session message when present for optimization
2. Validate that in-stroke curve data only appears with RecordingStrategy=StrokeBoundary (or Unknown)

3.3 Optional Behaviors

Consumers MAY:

1. Optimize parsing based on RecordingStrategy value (e.g., skip stroke detection in StrokeBoundary files)
2. Issue warnings when encountering unexpected combinations (e.g., GPS-update with in-stroke data)

4. Native FIT Field Mappings

Producers SHOULD use these native FIT fields for rowing data:

FIT Field	Type	Usage	Notes
timestamp	UINT32	Record timestamp	Milliseconds since Garmin epoch (1989-12-31 UTC)
distance	UINT32	Cumulative distance	Meters, scale 100
cadence	UINT8	Stroke rate	Strokes per minute
heart_rate	UINT8	Heart rate	Beats per minute, 0-255
power	UINT16	Average power	Watts, 0-65535
enhanced_speed	UINT32	Boat speed	Meters per second (from pace)
position_lat	SINT32	Latitude	Semicircles
position_long	SINT32	Longitude	Semicircles
total_cycles	UINT32	Cumulative stroke count	MAY repeat (GPS-update) or increment by >1

FIT Field	Type	Usage	Notes
cycle_length16	UINT16	Stroke distance	Distance per stroke cycle, scale 100, max 655m

5. Developer Field Specifications

5.1 Core Rowing Metrics (Record-Level)

Field Name	ID	Base Type	Scale	Units	Definition	Typical Range
DriveLength0		UINT16	100	m	Distance traveled by handle along longitudinal axis during drive phase	1.2-1.5 m
StrokeDriveTime	1	UINT16	1	ms	Duration of drive phase	300-600 ms
DragFactor 2		UINT16	1		Resistance setting (ergometer)	Device-specific
StrokeRecoveryTime	3	UINT16	1	ms	Duration of recovery phase	500-1500 ms
AverageDriveForce	4	UINT16	10	lbs	Average force during drive (deprecated)	-

Field Name	ID	Base Type	Scale	Units	Definition	Typical Range
PeakDriveForceLength	5	UINT16	10	lbs	Peak force during drive (deprecated)	-
AverageDriveForce	6	UINT16	10	N	Average force during drive phase	200-600 N
PeakDriveForceNU	7	UINT16	10	N	Peak force during drive phase	400-1200 N
AverageBoatSpeed	8	UINT16	100	m/s	Average boat speed during stroke	3-6 m/s
WorkoutState	9	UINT8	1		Rowing state indicator	See WorkoutState values
StrokeWork	19	UINT16	1	J	Work done over full stroke cycle	100-500 J

Notes:

- **DriveLength:** For OTW rowing, projection of handle trajectory on longitudinal axis. For indoor, handle travel catch-to-finish.
- **Force fields:** Newtons (IDs 6-7) are RECOMMENDED. Pounds (IDs 4-5) retained for backward compatibility only.
- **StrokeWork:** Energy over complete stroke cycle (not drive-only). Equivalent to average power × stroke period.

5.2 Oarlock Metrics (Single, Record-Level)

Field Name	ID	Base Type	Scale	Units	Definition
Catch	11	SINT16	10	deg	Oar angle at catch (0° = perpendicular to boat)
Finish	12	SINT16	10	deg	Oar angle at finish (0° = perpendicular to boat)
Slip	13	SINT16	10	deg	Oar slip angle (early blade entry)
Wash	14	SINT16	10	deg	Wash angle (late blade exit)
PeakForceAngle	15	SINT16	10	deg	Oar angle at peak force (0° = perpendicular to boat)
EffectiveLength	16	UINT16	100	m	Effective oar lever length (rigging)
PeakForcePositionNorm	17	SINT16	1		Normalized position of peak force along drive (0-10000)
PeakForcePositionAbs	18	UINT16	100	m	Absolute handle position at peak force

Notes:

- **Oar angles** (Catch, Finish, PeakForceAngle) use the rowing convention: **0° = oar perpendicular to the boat's longitudinal axis**. Negative values indicate the catch direction (oar blade toward bow, handle toward stern); positive values indicate the finish direction (oar blade toward stern, handle toward bow). This is standard oarlock/gateforce sensor convention.
- Summary fields (Catch, Finish, etc.) serve as representative values when per-side data unavailable
- **PeakForcePositionNorm**: Value in range 0-10000 representing ten-thousandths (0 = catch, 10000 = end of drive)

- **PeakForceAngle** vs **PeakForcePosition**: Angle is for oar angle sensors (OTW); Position is for handle position sensors (in-door)

5.3 Dual Oarlock Metrics (Record-Level)

When both port and starboard oarlocks are present, per-side metrics MAY be included:

Field Name	ID	Base Type	Scale	Units	Side
CatchPort	200	SINT16	10	deg	Port
CatchStarboard	201	SINT16	10	deg	Starboard
FinishPort	202	SINT16	10	deg	Port
FinishStarboard	203	SINT16	10	deg	Starboard
SlipPort	204	SINT16	10	deg	Port
SlipStarboard	205	SINT16	10	deg	Starboard
WashPort	206	SINT16	10	deg	Port
WashStarboard	207	SINT16	10	deg	Starboard
PeakForceAnglePort	208	SINT16	10	deg	Port
PeakForceAngleStarboard	209	SINT16	10	deg	Starboard
EffectiveLengthPort	210	UINT16	100	m	Port
EffectiveLengthStarboard	211	UINT16	100	m	Starboard

Per-side field rules:

- Per-side fields SHOULD only be included when both port and starboard data are available
- When per-side fields are present, summary fields (IDs 11-16) SHOULD contain the average of port and starboard values
- When only one side is available, summary fields SHOULD contain that side's value
- Consumers implementing only partial support MAY ignore per-side fields and use summary fields

6. In-Stroke Curve Data

6.1 Overview

In-stroke curve data provides high-resolution force, acceleration, or angle measurements throughout a stroke cycle. This data enables detailed stroke analysis and technique visualization.

Constraint: In-stroke curve data MUST only appear when RecordingStrategy=StrokeBoundary (or Unknown with stroke-boundary semantics), as curves require stroke boundaries for interpretation.

6.2 Field ID Allocation

- **90-92**: Axis metadata (per-Record)
- **20-59**: Curve summary statistics (dynamic, per-curve-type)
- **60-255**: Curve arrays (dynamic, per-curve-type)

6.3 Axis Metadata Fields (Record-Level)

These fields define the X-axis interpretation for curve data on each Record:

Field Name	ID	Base Type	Scale	Units	Definition
InstrokeAbscissaType	90	UINT8	1		X-axis semantics (enum)
InstrokeSampleInterval	91	UINT16	1	(varies)	Sample spacing (interpretation depends on Type)
InstrokePointCount	92	UINT8	1		Number of points in curve arrays

InstrokeAbscissaType values:

Value	Name	InstrokeSampleInterval Meaning
0	UNKNOWN	Not specified (shape-only curves)
1	TIME_UNIFORM_MS	Time in ms between samples
2	HANDLE_DISTANCE_UNIFORM_M	Distance between samples (scale as documented)
3	OAR_ANGLE_UNIFORM_DEG	Degrees between samples (scale as documented)
4	NORMALIZED_INTERVALS	Time in seconds step size (0-1 range)

Rules:

- Axis metadata fields **MUST** appear together on each Record containing curve data
- For Type=TIME_UNIFORM_MS with known drive time: $\text{InstrokeSampleInterval} = \text{drive_time_ms} / (\text{point_count} - 1)$
- For Type=OAR_ANGLE_UNIFORM_DEG: Domain is [Catch, Finish] angles (from fields 11-12)

- Type=UNKNOWN indicates shape-only data (for pattern analysis, not absolute plotting)
- Producers SHOULD strive for consistency between axis meta-data and stroke scalars, but consumers SHOULD NOT enforce strict validation

6.4 Standard Curve Types

Recommended curve type names:

Curve Name	Typical Data	Units
HandleForceCurve	Handle force over stroke	Newtons
BoatAcceleratorCurve	Boat acceleration	m/s ²
OarAngleVelocityCurve	Angular velocity of oar	deg/s
SeatCurve	Seat position	meters

6.5 Curve Array Format

Curve data MUST be encoded as **UINT16** arrays (developer fields with array size > 1):

- **Maximum points per curve:** 127 (FIT limit: 255 bytes / 2 bytes per UINT16)
- **Field ID allocation:** Start at 60, increment per curve type
- **Encoding:** Unsigned 16-bit integers in range [0, 65535]
- **Data representation:** Values are clipped to [0, 65535] range; negative values not supported
- **Scale factor:** Use appropriate scale in field description to map physical units to UINT16 range

Note on signed data: FIT SDK developer fields with arrays only reliably support UINT16 base type. For force curves and other rowing metrics, values are naturally non-negative. If future curve types require negative values, implement offset transformation (e.g., add 32768) and document in field description.

6.6 Curve Summary Statistics

As an alternative or supplement to full curves, summary statistics MAY be provided:

- **Field ID allocation:** Start at 20, increment per curve and metric
- **Metrics:** q1, q2, q3, q4 (quartile variations), diff (change), maxpos, minpos (normalized positions)

6.7 Companion Files

For curves exceeding 127 points, producers MAY use companion JSON files:

- **Filename:** Same basename as FIT file with `.instroke.json` extension
- **Format:** JSON object with curve names as keys, arrays of per-stroke samples as values
- **Metadata:** Include `_rowingdata_instroke` object with version, abscissa type, point counts

7. Compliance Levels

Implementers MAY choose compliance levels based on their capabilities:

Level 1: Minimal

- Native FIT fields only
- `RecordingStrategy` on Session (recommended)
- Supports at least one recording strategy

Level 2: Standard

- Level 1 requirements
- Core rowing metrics (IDs 0-9, 19)
- Handles missing developer fields gracefully

Level 3: Full

- Level 2 requirements
- Oarlock metrics (IDs 11-18)
- In-stroke axis metadata (IDs 90-92)
- At least one in-stroke curve type

Level 4: Advanced

- Level 3 requirements
- Dual oarlock per-side metrics (IDs 200-211)
- Multiple in-stroke curve types
- Curve summary statistics

8. Data Quality and Validation

8.1 Value Ranges

Producers SHOULD clamp values to reasonable ranges:

- Force: 0-2000 N typical maximum
- DriveLength: 0.8-2.0 m
- Angles: -180 to +180 degrees
- Stroke rate: 10-60 strokes/min typical

8.2 Missing Data

- Missing or unavailable fields SHOULD be omitted from the file
- Producers MUST NOT emit fields with placeholder values (e.g., -1, 999) without documentation
- Zero values SHOULD indicate actual measurements of zero (not missing data)

8.3 Consistency

While strict validation is not enforced, producers SHOULD maintain internal consistency:

- Drive + recovery time $\approx$ stroke period (60 / cadence)
- In-stroke axis metadata consistent with stroke scalars
- Summary oarlock fields = average of per-side fields (when both present)

9. Field Registry

9.1 Reserved ID Ranges

Range	Purpose	Status
0-19	Core rowing metrics	Assigned
20-59	In-stroke summaries	Dynamic allocation
60-89	In-stroke curve arrays	Dynamic allocation
90-92	In-stroke axis metadata	Assigned
93-199	Reserved for future standard fields	Available
200-211	Dual oarlock per-side	Assigned
212-255	Reserved for future extensions	Available

9.2 Deprecated Fields

Field ID	Name	Status	Replacement
4	AverageDriveForceLbs	Deprecated	AverageDriveForceN (ID 6)
5	PeakDriveForceLbs	Deprecated	PeakDriveForceN (ID 7)

Producers SHOULD use Newtons for new implementations. Consumers MUST continue to support pounds for backward compatibility.

10. Version History

Version	Date	Changes
1.1	2026-05-14	FIT SDK compliance updates: Application ID changed to 16-byte UUID (74c89dce-0f16-5d5a-aeb1-6c9f3e6d8f3a); Curve arrays changed from SINT16 to UINT16 for developer field compatibility
1.0	2026-05-13	Initial standard release

Appendix A: Terminology

Stroke Cycle: Complete rowing motion from catch through drive and recovery back to catch.

Drive Phase: Portion of stroke where power is applied (handle moving toward finish).

Recovery Phase: Portion of stroke returning to catch position (no power).

Catch: Position/angle at start of drive phase (blade entry for OTW).

Finish: Position/angle at end of drive phase (blade exit for OTW).

Slip: Angular difference between ideal and actual blade entry (early entry).

Wash: Angular difference between ideal and actual blade exit (late exit).

Oar Angle: Angle of oar relative to boat's longitudinal axis (0° = perpendicular).

Effective Length: Horizontal distance from oarlock pin to handle (rigging metric).

Drive Length: Actual distance handle travels during drive phase.

Stroke Distance: Distance boat travels during one complete stroke cycle.

OTW: On-the-water (as opposed to ergometer/indoor).

Appendix B: Reference Implementation

A reference implementation of this standard is available in the **rowingdata** open-source Python library:

- **Repository:** <https://github.com/sanderroosendaal/rowingdata>
- **Machine-readable specification:** `rowingdata/data/fit_export_spec.json`
- **Example FIT file:** `testdata/rowingdata_standard_example.fit`

Appendix C: Acknowledgments

This standard builds upon work by the rowing data community, including contributions from:

- Device manufacturers (Concept2, Nielsen-Kellerman, Quiske, RowPerfect)
- Software platforms (Intervals.icu, OpenRowingMonitor)
- Individual developers and researchers

Appendix D: Contact and Governance

For questions, clarifications, or proposed amendments to this standard:

- **Issue tracker:** <https://github.com/sanderroosendaal/rowingdata/issues>
- **Discussion forum:** Rowing Industry Trade Association meetings

Proposed changes SHOULD be discussed with stakeholders before implementation to ensure ecosystem compatibility.

End of Standard